QUESTION 2 · TECHNICAL BRIEF

ESG Key Performance Indicators for a Refinery under GRI 11

How the Global Reporting Initiative's oil and gas sector standard works, which of its 22 topics a refinery must answer for, and the exact metrics, units and formulas that go in the sustainability report.

Group C (29 Members)

Abu Omosomi Joy · Iniovosa Michayah Olamide

Abdulkadir Garba Yusuf Aondoakaa · Doochivir Christopher

01 Why a refinery reports

A sustainability report is a public account of your effect on the economy, the environment and people. In Nigeria, several separate obligations now converge on it.

ESG stands for **Environmental, Social and Governance** — the three families of non-financial performance that investors, regulators and communities now expect to be measured with the same discipline as profit. A **KPI**, or **Key Performance Indicator**, is a single defined number that tracks one of them over time.

Nigerian requirement	What it obliges a refinery to disclose
Climate Change Act 2021	**Section 24**: entities with 50 or more employees appoint a Climate Change Officer and file **annual carbon emissions and reduction reports**. This is the statutory floor beneath the climate section of any report.
NGX Sustainability Disclosure Guidelines	The **Nigerian Exchange** expects listed companies to publish sustainability information, and points explicitly at the **GRI Standards** as the reference framework.
FRC of Nigeria	The **Financial Reporting Council of Nigeria** has adopted **IFRS S1 and S2**, the **International Sustainability Standards Board**'s disclosure standards, on a phased roadmap. Nigeria was among the earliest jurisdictions to commit.
Nigerian Code of Corporate Governance 2018	Requires the board to report on sustainability policy and performance, including environmental and community matters.
NEITI Act 2007	The **Nigeria Extractive Industries Transparency Initiative** reconciles company payments against government receipts and publishes the result — a statutory transparency duty that maps directly onto GRI topic 11.21.
Nigerian Content Act 2010	Administered by the **Nigerian Content Development and Monitoring Board (NCDMB)**. Local employment, local procurement and local capacity building are legal obligations here, not voluntary good practice.
PIA 2021, NMDPRA, NESREA, NOSDRA	Licence conditions, air and effluent standards, spill reporting. The **Petroleum Industry Act**'s host community provisions have also reset what communities expect of any petroleum facility.

IN PLAIN TERMS

Financial statements tell shareholders how much money the business made. A sustainability report tells everyone else what the business did to the air, the water, the workforce and the community while making it. **GRI is the grammar for writing that second account so that it can be compared with anyone else's.**

THE DISTINCTION THAT TRIPS PEOPLE UP

The **GHG inventory** from Question 1 is an *input*. The **sustainability report** is the *output*. The inventory produces the tonnes; GRI decides which tonnes get published, alongside what else, in what units, with what context. **If the two use different boundaries, that difference must be stated explicitly** — an unexplained mismatch is the single fastest way to lose an assurance opinion.

02 How GRI fits together

Three layers of standards. You use all three, in order, and the sector layer is what makes the report specific to refining.

Layer	Standards	What it does
1. Universal — everyone	**GRI 1** Foundation 2021 **GRI 2** General Disclosures 2021 **GRI 3** Material Topics 2021	GRI 1 sets the rules of use and the reporting principles. GRI 2 covers who you are: governance, strategy, policies, workforce, value chain. GRI 3 is the **process for deciding what matters**.
2. Sector — your industry	**GRI 11** Oil and Gas Sector 2021 effective 1 January 2023	Lists **22 topics that are likely to be material** for any oil and gas organisation, and for each one points to the disclosures you should report. GRI 11 was the **first sector standard GRI ever published**.
3. Topic — the metrics	**GRI 200** Economic **GRI 300** Environmental **GRI 400** Social	The actual disclosures and their definitions — for example **GRI 305-1** Direct (Scope 1) GHG emissions, or **GRI 403-9** Work-related injuries.

The sequence you actually follow

Step	What you do	Standard used
1	Understand your context, activities and business relationships.	GRI 3-1
2	Identify your actual and potential impacts on the economy, environment and people.	GRI 3-1, **plus the 22 topics of GRI 11 as your starting checklist**
3	Assess significance and prioritise. Produce your list of material topics.	GRI 3-2
4	For every material topic, explain how you manage it, then report the mapped disclosures.	GRI 3-3 + Topic Standards
5	**For every GRI 11 topic you concluded is not material, say so and explain why.**	GRI 3-2, GRI 11
6	Publish a GRI content index mapping every disclosure to where it is answered.	GRI 1, GRI 2-3

THE REPORT-OR-EXPLAIN RULE — THE HEART OF GRI 11

You do not get to quietly skip a topic. GRI 11 names 22 topics as likely material for this sector. If you decide one of them does not apply to you, **you must list it and give the reason**. That single mechanism is what stops a refinery publishing a report about tree planting and community football while saying nothing about flaring, process safety or payments to governments.

A NOTE ON DISCLOSURE NUMBERING

GRI 11 numbers its disclosures sequentially inside each topic — so under topic 11.1 GHG emissions, **11.1.2** points to GRI 302-1, **11.1.5** points to GRI 305-1, and so on. Because the pointers matter less than the underlying metric, **this brief cites the Topic Standard codes** (305-1, 403-9, 207-4), which are stable and unambiguous. Confirm the 11.x.y cross-references against the official GRI 11 PDF when you build your content index.

03 Three kinds of materiality

The most misunderstood word in sustainability reporting. Three major frameworks use it to mean three different things, and mixing them produces an incoherent report.

Framework	Materiality lens	The question it asks	Consequence
GRI	**Impact materiality** inside → out	What are our most significant effects **on** the economy, environment and people?	A community water impact is material even if it never costs the company a naira.
ISSB IFRS S1 & S2	**Financial materiality** outside → in	Which sustainability matters could reasonably affect our **cash flows, access to finance or cost of capital**?	Aimed at investors. A physical climate risk to the jetty is material; a diffuse social impact may not be.
ESRS under the EU CSRD	**Double materiality** both directions	Either lens qualifying makes the topic material.	The widest net. Required of firms in scope of the EU **Corporate Sustainability Reporting Directive**.

IN PLAIN TERMS

GRI asks **"what are we doing to the world?"** The ISSB asks **"what is the world going to do to us?"** The EU asks both. A Nigerian refinery reporting under GRI while also answering the FRC's IFRS S2 requirement is effectively doing double materiality — so **run one assessment that captures both lenses and tag each topic with which lens it qualifies under.** Doing two separate assessments produces two lists that will not reconcile, and reviewers notice.

Running the assessment

Stage	What good looks like at a refinery
Identify impacts	Start from the 22 GRI 11 topics. Add anything specific to your site: a nearby fishing community, an abstraction licence, a legacy contamination liability.
Engage stakeholders	Host communities, workforce and contractors, regulators (NMDPRA, NESREA, NOSDRA), lenders, offtakers, host government, civil society. **Record who you spoke to and what they said** — GRI 2-29 asks for exactly this.
Assess significance	For actual impacts, severity (scale, scope, remediability). For potential impacts, severity **and** likelihood. For human rights impacts, **severity outweighs likelihood**.
Prioritise	Rank and set a cut-off. Have the board or an executive committee approve the list — GRI 3-1 expects governance involvement, not a consultant's spreadsheet.
Review	Refresh annually; re-run fully every two to three years or after any material change to the business.

THE SEVERITY TEST, IN ONE LINE

Severity is not the same as likelihood, and for anything touching human rights it wins. **A rare event that could displace a community is material. A frequent event that annoys nobody is not.**

04 The 22 topics of GRI 11

Every one must be reviewed and either reported or explained away. The right-hand column is the typical verdict for a downstream refinery.

Topic	Name	What it means at a refinery	Typical verdict
11.1	**GHG emissions**	Scope 1, 2 and 3; energy use and intensity; methane; flaring and venting.	**Material**
11.2	**Climate adaptation, resilience and transition**	Physical risk to the site; transition risk to demand; scenario analysis.	**Material**
11.3	**Air emissions**	Sulphur dioxide, nitrogen oxides, particulates, volatile organic compounds, benzene.	**Material**
11.4	**Biodiversity**	Habitat at the site, jetty and pipeline corridors; coastal and wetland receptors.	Site specific
11.5	**Waste**	Spent catalyst, oily sludge, tank bottoms, spent caustic, contaminated soil.	**Material**
11.6	**Water and effluents**	Withdrawal, consumption, discharge quality, water stress in the catchment.	**Material**
11.7	**Closure and rehabilitation**	Decommissioning liability, provisioning, land restoration.	**Material**
11.8	**Asset integrity and critical incident management**	Process safety events, loss of primary containment, major accident hazard control.	**Material**
11.9	**Occupational health and safety**	Injuries, ill health, contractor safety, occupational exposure to benzene and H_2S.	**Material**
11.10	**Employment practices**	Terms, benefits, turnover, contractor and migrant labour conditions.	**Material**
11.11	**Non-discrimination and equal opportunity**	Pay equity, representation, discrimination incidents and remedy.	**Material**
11.12	**Forced labour and modern slavery**	Risk concentrated in construction, security and catering contractor tiers.	Supply chain
11.13	**Freedom of association and collective bargaining**	Union recognition, collective agreements, contractor rights.	**Material**
11.14	**Economic impacts**	Local employment and procurement, Nigerian content, taxes.	**Material**
11.15	**Local communities**	Host community engagement, grievance mechanism, social investment.	**Material**
11.16	**Land and resource rights**	Acquisition, compensation, resettlement, access to land and water.	Site specific
11.17	**Rights of indigenous peoples**	Free, prior and informed consent where indigenous groups are present.	Explain if not
11.18	**Conflict and security**	Security arrangements and human rights; operating in conflict-affected areas.	**Material**
11.19	**Anti-competitive behaviour**	Pricing, market allocation, product supply conduct.	**Material**
11.20	**Anti-corruption**	Bribery risk in permitting, customs, procurement and logistics.	**Material**
11.21	**Payments to governments**	Taxes, royalties, levies and fees by country and by project.	**Material**
11.22	**Public policy**	Lobbying, political contributions, trade association positions on climate.	**Material**

Verdicts are typical for a downstream refinery and illustrative only; your own GRI 3 assessment decides.

05 KPIs: climate and energy

Topics 11.1 and 11.2. These carry the most scrutiny and are where the GHG inventory from Question 1 is published.

GRI	Key performance indicator	Unit	Definition, formula or note
305-1	Gross direct (Scope 1) GHG emissions	tCO_2e	All seven gases, gross — before any offset or credit. State the GWP source (AR6) and the consolidation approach.
305-2	Energy indirect (Scope 2) emissions	tCO_2e	**Report both location-based and market-based.** Neither substitutes for the other.
305-3	Other indirect (Scope 3) emissions	tCO_2e	By category. **Category 11, use of sold products, is the one that must not be omitted** — it is around 87% of a refinery's footprint.
305-4	GHG emissions intensity	tCO_2e / t crude tCO_2e / CWT	Emissions ÷ throughput. State the numerator scopes. **CWT** (Complexity Weighted Tonne) allows fair comparison between refineries of different configuration.
305-5	Reduction of GHG emissions	tCO_2e	Reductions achieved against the base year, by initiative. Distinguish genuine abatement from throughput or divestment effects.
Sector	Methane emissions	t CH_4 %	Methane in tonnes, not only CO_2e — the CO_2e conversion hides the volume. **Methane intensity** = methane emitted ÷ gas throughput.
Sector	Hydrocarbons flared and vented	million Nm^3 tCO_2e	Split **routine** from **non-routine** (upset, start-up, emergency). Routine flaring is the number regulators and investors track.
302-1	Energy consumption within the organisation	GJ	Fuel plus purchased electricity, heat, steam and cooling, less any exported.
302-3	Energy intensity	GJ / t crude EII index	The industry benchmark is the **Solomon Energy Intensity Index**: actual energy against a standard for your configuration, where 100 is par.
302-4	Reduction of energy consumption	GJ	Savings delivered by named initiatives, against a stated baseline.
201-2	Financial implications of climate change	Narrative + ₦	Risks and opportunities, their financial effect, and the cost of managing them. This is where GRI overlaps IFRS S2 most directly.
Sector	Transition plan and scenario analysis	Narrative	Which scenarios were tested (for example an International Energy Agency net-zero pathway), the resilience conclusion, and **the share of capital expenditure aligned to low carbon**.

THREE TESTS A REVIEWER WILL APPLY IMMEDIATELY

1. Does Scope 3 Category 11 appear? Its absence is read as concealment. **2. Are targets absolute or only intensity?** Intensity can fall while absolute emissions rise, so an intensity-only target is treated as weak. **3. Do the numbers reconcile with the GHG inventory and with any CDP submission?** Three different figures for the same year destroys confidence in all three.

06 KPIs: air, water, waste and land

Topics 11.3 to 11.7. These are the impacts a host community can see, smell and taste, which is why they carry disproportionate reputational weight.

11.3 AIR EMISSIONS

GRI	Key performance indicator	Unit	Definition, formula or note
305-7	SO_2, NO_x, particulates, VOC	tonnes/yr	Sulphur dioxide, nitrogen oxides, particulate matter, volatile organic compounds. Report method: continuous monitoring, stack testing or emission factor.
305-7	Sulphur recovery efficiency	%	Sulphur recovered ÷ sulphur in feed to the recovery unit. A modern Claus plant with tail gas treatment should exceed 99.5%.
305-7	Benzene at the fenceline	µg/m³	Passive sampler ring around the boundary. The recognised benchmark action level is **9 µg/m³** annual average, from the US refinery sector rule.
305-6	Ozone-depleting substances	t CFC-11e	Refrigerant losses, expressed as CFC-11 equivalent.

11.6 WATER AND EFFLUENTS

GRI	Key performance indicator	Unit	Definition, formula or note
303-3	Water withdrawal by source	megalitres	Surface, ground, sea, produced and third-party water. **Report separately the share drawn from water-stressed areas.**
303-4	Water discharge and quality	ML; mg/L	Volume by destination, plus oil in water, chemical oxygen demand, phenol, ammonia and suspended solids.
303-5	Water consumption	megalitres	Withdrawal minus discharge — the water that does not come back, mostly evaporative loss from cooling.
303-2	Discharge quality standards applied	Narrative	The limits you hold yourself to where no local standard exists.

11.5 WASTE · 11.4 BIODIVERSITY · 11.7 CLOSURE

GRI	Key performance indicator	Unit	Definition, formula or note
306-3	Waste generated, hazardous and non-hazardous	tonnes	Refinery-specific streams: spent catalyst, oily sludge, tank bottoms, spent caustic, contaminated soil.
306-4	Waste diverted from disposal	tonnes; %	Reuse, recycling and recovery. Spent catalyst metal reclamation is the largest single opportunity at most refineries.
306-5	Waste directed to disposal	tonnes	By method: incineration, landfill, other. State whether on-site or off-site.
306-1/2	Significant spills	count; m³	Number and volume above the reporting threshold, with volume recovered. In Nigeria these are also reportable to **NOSDRA**.
304-1/2	Sites in or adjacent to protected areas; habitat affected	count; ha	Include the jetty, pipeline corridors and abstraction points, not only the plot fence.
Sector	Closure and rehabilitation provision	₦ million	The financial provision recognised for decommissioning and site restoration, and how it is secured.

WHAT AN OPERATOR DOES

Tie each KPI to a system that already exists — air to the NESREA returns, water to the effluent plant laboratory record, waste to the manifest system, spills to the NOSDRA log. **A KPI with no system underneath it gets estimated at year end, and it shows.**

07 KPIs: safety and workforce

Topics 11.8 to 11.13. Topic 11.8 is where a refinery's process safety record becomes a public disclosure.

11.8 ASSET INTEGRITY AND CRITICAL INCIDENT MANAGEMENT

GRI	Key performance indicator	Unit	Definition, formula or note
Sector	**Tier 1 process safety events**	count	Loss of primary containment above defined severity thresholds, per **API RP 754** / IOGP 456. The headline process safety number in this industry.
Sector	Tier 2 process safety events	count	Lesser but still significant releases. Tier 3 and 4 are leading indicators, usually managed internally.
Sector	Process safety event rate	per million hours	**(Tier 1 events × 1,000,000) ÷ hours worked**, including contractors.
Sector	Loss of primary containment count	count	All unplanned releases, including those below Tier 2.
3-3	Asset integrity management approach	Narrative	Risk-based inspection, integrity operating windows, safety-critical element assurance, management of change, permit to work, emergency response.

11.9 OCCUPATIONAL HEALTH AND SAFETY

GRI	Key performance indicator	Unit	Definition, formula or note
403-9	Fatalities from work-related injury	count; rate	Employees and workers who are not employees but whose work is controlled by the organisation, reported separately.
403-9	High-consequence work-related injuries	count; rate	Injuries from which the worker cannot fully recover within six months.
403-9	Total recordable injury rate	per 200,000 h or per 1,000,000 h	**(Recordable cases × 200,000) ÷ hours worked.** The 200,000 basis is US practice; much of the industry uses 1,000,000. **State the basis** — they differ fivefold.
403-10	Work-related ill health	count	Occupational illness, including benzene and noise exposure cases.
403-8	Workers covered by the OH&S management system	%	Say whether the system is certified, for example to **ISO 45001**.
403-1..7	System, hazard identification, training, participation	Narrative	Including the right to remove oneself from unsafe work — a disclosure reviewers look for specifically.

11.10 TO 11.13 WORKFORCE

GRI	Key performance indicator	Unit	Definition, formula or note
2-7 / 2-8	Employees and non-employee workers	headcount	By contract, gender and region. **Contractors often outnumber employees**, and excluding them distorts every rate above.
401-1	New hires and turnover	count; %	By age group, gender and region.
405-1	Diversity of governance bodies and employees	%	By gender and age band, at board, management and workforce level.
405-2	Ratio of basic salary, women to men	ratio	By employee category and location.
406-1	Incidents of discrimination	count	And the remediation taken for each.
407-1 / 409-1	Freedom of association and forced labour risk	count	Operations and suppliers at risk. **Concentrate on construction, security and catering contractors**, where sector risk is highest.
2-30	Collective bargaining coverage	%	Share of employees covered by collective agreements.

08 KPIs: society and communities

Topics 11.14 to 11.18. In Nigeria several of these are backed by their own statute, which raises the evidential bar.

11.14 ECONOMIC IMPACTS · 11.15 LOCAL COMMUNITIES · 11.16 TO 11.18 RIGHTS AND SECURITY

GRI	Key performance indicator	Unit	Definition, formula or note
204-1	Proportion of spending on local suppliers	%	Local procurement ÷ total procurement. In Nigeria this is also a **Nigerian Content Act** obligation reported to the NCDMB — use the same figure in both.
202-2	Senior management hired from the local community	%	Define "local" explicitly: host community, state, or nationally.
202-1	Entry level wage against local minimum wage	ratio	By gender, at significant locations.
203-1	Infrastructure investments and services supported	₦ million	Roads, water, power, schools and clinics — and whether commercial or pro bono.
203-2	Significant indirect economic impacts	Narrative	Jobs induced in the local economy, supplier development, skills transfer.
413-1	Operations with community engagement and impact programmes	%	Target is 100% of sites, with a documented engagement plan for each.
413-2	Operations with significant negative community impacts	count	Actual and potential. Name them and state the mitigation.
2-25 / 2-26	Community grievances	count; days	**Received, resolved, and average days to close.** The closure time is the number that shows whether the mechanism is real.
411-1	Violations of the rights of indigenous peoples	count	Where indigenous groups are present. If none, say so and explain the basis.
Sector	Resettlement and land access	households	Households economically or physically displaced, and compensation paid. Benchmark against **IFC Performance Standard 5**.
410-1	Security personnel trained in human rights	%	Including third-party guards and any state security. The reference framework is the **Voluntary Principles on Security and Human Rights**.

WHAT AN OPERATOR DOES

Run the community grievance mechanism as a logged, time-bound process with the same discipline as a maintenance work order: every grievance numbered, owned, and closed against a target. **The number that proves it works is not how many you received — it is how long they take to close.**

09 KPIs: governance and transparency

Topics 11.19 to 11.22. In Nigeria two of these are backed by independent statutory reconciliation, which raises the evidential bar and the opportunity.

THE DISCLOSURES

GRI	Key performance indicator	Unit	Definition, formula or note
205-1	Operations assessed for corruption risk	%; count	And the significant risks identified. Focus on permitting, customs, product allocation and logistics.
205-2	Anti-corruption policy communication and training	%	By employee category and region, **and for business partners** — the part most reports omit.
205-3	Confirmed incidents of corruption	count	And the action taken, including dismissals and contracts terminated.
206-1	Legal actions for anti-competitive behaviour	count	Anti-trust and monopoly actions, and their outcomes.
207-1..3	Tax approach, governance and stakeholder engagement	Narrative	Tax strategy, who owns it, and how tax concerns are handled.
207-4	**Country-by-country reporting**	₦ million	Revenue, profit, tax paid and accrued, employees and tangible assets, per jurisdiction.
Sector	Payments to governments	₦ million	By type — taxes, royalties, levies, licence fees — and **by project**. Should reconcile to the **NEITI** audit; a difference you cannot explain is a finding.
415-1	Political contributions	₦	Direct and indirect, by recipient.
2-28	Membership associations	list	**And whether their climate positions align with yours.** Misalignment between a company's stated climate policy and its trade association's lobbying is now a standard analyst challenge.

THE NIGERIAN EVIDENTIAL ADVANTAGE

Because NEITI already reconciles payments, and the NCDMB already receives local content returns, a Nigerian refinery has **independently verified numbers** available for two of the topics that companies elsewhere find hardest to evidence. Use them. Citing a NEITI-reconciled figure is far stronger than a self-declared one.

10 Industry alternatives to GRI

GRI is not the only framework, and most refiners report against several at once. Each answers a different question for a different reader.

Framework	Who it serves	Materiality	What it adds, and when you would use it
GRI + GRI 11	All stakeholders	Impact	The broadest and most recognised. The default backbone, and the one NGX points to.
IFRS S1 & S2 (ISSB)	Investors	Financial	Governance, strategy, risk management, metrics and targets, with mandatory scenario analysis. **Adopted by the FRC of Nigeria**, so it is becoming compulsory. Absorbed the TCFD recommendations.
SASB standards (now within ISSB)	Investors	Financial	Industry-specific metrics. The relevant standard is **Oil & Gas — Refining & Marketing**: precise, and a good pair with GRI's breadth.
IPIECA Sustainability Reporting Guidance, 4th edition	The sector itself	Impact	Written by the industry for the industry and **already cross-mapped to GRI**. The most practical companion when deciding how to measure a sector indicator.
CDP	Investors and buyers	Financial	An annual questionnaire on climate, water and forests, scored A to D−. Not a standard, but a benchmark customers and lenders read.
ESRS under the EU CSRD	EU regulators	Double	Prescriptive and extensive. Relevant with EU-listed parentage or significant EU turnover.
OGMP 2.0	Regulators, buyers	—	United Nations methane framework, Levels 1 to 5. **Gold Standard** requires measurement-based reporting — the strongest evidence for the methane KPI.
EITI / NEITI	Citizens	—	Independent reconciliation of payments to governments. Statutory in Nigeria; the natural evidence base for topic 11.21.
UN SDGs	General public	—	Seventeen Sustainable Development Goals. Useful for framing; **not a measurement standard**, and no substitute for one.

HOW THESE ARE ACTUALLY COMBINED

The common architecture is **GRI as the backbone, IFRS S2 for the investor-facing climate section, SASB metrics where a precise industry number is wanted, and IPIECA for how to measure sector-specific indicators**; CDP is then answered from the same data set. What makes it work is a **single data dictionary** — one definition and one owner per metric, mapped to every framework that asks for it. Teams that keep a separate spreadsheet per framework publish contradictory numbers, and it is always noticed.

GRI tells the world what you did to it; the ISSB tells investors what it might cost you. **Different readers, same plant.**

11 Assurance, pitfalls and the cycle

A report nobody can verify is marketing. These are the checks, the traps, and the calendar.

Getting it assured

Element	What to do
Standard	**ISAE 3000 (Revised)** for non-financial information generally, and **ISAE 3410** for greenhouse gas statements specifically. **AA1000AS** is the alternative where stakeholder responsiveness is to be assessed as well.
Level	**Limited** assurance is the normal starting point; **reasonable** assurance is increasingly expected for Scope 1 and 2 and for safety data. State clearly which applies to which KPI — scope creep in the assurance statement is a common criticism.
Scope	Name exactly which indicators were assured. A statement covering only three KPIs, presented as though it covers the report, is misleading.
Restatement	Publish a restatement policy. When a prior-year figure changes, show the old value, the new value and the reason.

The pitfalls that damage credibility

Pitfall	Why it fails, and what to do instead
Omitting Scope 3 Category 11	Removes roughly nine tenths of the footprint. Disclose it, even if it is uncomfortable.
Boundary mismatch	The GHG inventory consolidates on operational control, the report quietly uses equity share. Align them, or state the difference and reconcile it.
Excluding contractors	Safety rates calculated on employee hours only, at a site where contractors are the majority of hours. Every rate becomes meaningless.
Undeclared rate basis	An injury rate per 200,000 hours compared against a peer's per 1,000,000 hours is a fivefold error. Always state the basis.
Selective baselines	Choosing a base year that happens to be an outlier. Disclose the basis and stick to it.
Silent omission	Dropping a GRI 11 topic without explanation. This breaches the report-or-explain rule directly.
Narrative without numbers	Photographs and case studies where a KPI belongs. Every material topic needs at least one measured indicator with a trend.
Association misalignment	Publishing a climate commitment while funding trade associations that lobby against it. Disclose memberships and address the gap.

The annual cycle

When	What happens
Q3, prior year	Refresh the materiality assessment. Confirm the KPI set and the data dictionary. Agree the assurance scope.
Q4	Collect and validate data monthly rather than annually. Close any gaps in the underlying operational systems.
Q1	Consolidate, calculate, and review against prior year. Investigate every unexplained variance before it reaches the auditor.
Q2	Independent assurance. Publish the report and the GRI content index. File under Climate Change Act section 24; submit NCDMB and NEITI returns from the same figures.

THE FINDING TO TAKE AWAY

GRI 11 does not simply supply a list of metrics. It supplies **a list of questions a refinery is not allowed to ignore**. The 22 topics are the sector's known impacts, assembled in one place, and the report-or-explain rule converts them from suggestions into obligations. **The quality of a sustainability report is decided by what it refuses to leave out.**

12 Reference

Every abbreviation used in this brief, and the standards it cites.

ABBREVIATIONS

AA1000AS	AccountAbility Assurance Standard
CDP	Global environmental disclosure platform (formerly the Carbon Disclosure Project)
CSRD	Corporate Sustainability Reporting Directive (European Union)
EITI	Extractive Industries Transparency Initiative
ESG	Environmental, Social and Governance
ESRS	European Sustainability Reporting Standards
FRC	Financial Reporting Council of Nigeria
GRI	Global Reporting Initiative
IFC	International Finance Corporation
IFRS	International Financial Reporting Standards
IPIECA	The global oil and gas industry association for environmental and social issues
ISAE	International Standard on Assurance Engagements
ISSB	International Sustainability Standards Board
KPI	Key Performance Indicator
NCDMB	Nigerian Content Development and Monitoring Board
NEITI	Nigeria Extractive Industries Transparency Initiative
NESREA	National Environmental Standards and Regulations Enforcement Agency
NGX	Nigerian Exchange
NMDPRA	Nigerian Midstream and Downstream Petroleum Regulatory Authority
NOSDRA	National Oil Spill Detection and Response Agency
OGMP	Oil and Gas Methane Partnership
PIA	Petroleum Industry Act 2021
SASB	Sustainability Accounting Standards Board
TCFD	Task Force on Climate-related Financial Disclosures
VOC	Volatile organic compounds

STANDARDS AND INSTRUMENTS CITED

GRI 1, 2, 3	Universal Standards 2021 — Foundation, General Disclosures, Material Topics
GRI 11	Oil and Gas Sector 2021 — effective 1 January 2023
GRI 200/300/400	Topic Standards — Economic, Environmental, Social
IFRS S1 & S2	General sustainability and climate-related disclosures (ISSB)
API RP 754	Process safety performance indicators for refining and petrochemicals
ISO 45001	Occupational health and safety management systems
ISAE 3000 / 3410	Assurance on non-financial information, and on greenhouse gas statements
IFC PS 5	Land acquisition and involuntary resettlement
Climate Change Act 2021	Nigeria — s.24 annual emissions reporting duty
Nigerian Content Act 2010	Nigeria — local employment, procurement and capacity building

BEFORE YOU SUBMIT

This brief cites GRI **Topic Standard** codes (305-1, 403-9, 207-4), which are stable and unambiguous. GRI 11 also numbers its own pointers sequentially within each topic — 11.1.2 for GRI 302-1, 11.1.5 for GRI 305-1. **Confirm those 11.x.y cross-references against the official GRI 11 PDF when you build your content index.**